

Grade 5 Cambridge Primary Science - Stage 6

Unit 2 Study Guide

2.1 — PROPERTIES OF SUBSTANCES

Name: _____

Class: _____

Date: _____

1. What is a substance?

A substance is a type of material.

Different substances have different properties.

A property tells us what a substance is like or how it behaves.

Examples of properties

A substance can be:

hard or soft

transparent or opaque

a good conductor or a poor conductor

magnetic or non-magnetic

soluble or insoluble

a solid, liquid or gas

2. States of matter

There are three common states of matter:

SOLID

A solid has a fixed shape and fixed volume.

Examples: Wood, metal

LIQUID

A liquid has a fixed volume but takes the shape of its container.

Examples: Oil, milk

GAS

A gas does not have a fixed shape or fixed volume.

Examples: Oxygen, carbon dioxide, water vapour

Gas Properties

Gas has MASS and TAKES UP SPACE

Remember

Solid → Liquid → Gas

When a substance is heated, it can change state. When it is cooled down it also can change state.

3. Melting and boiling

Melting- happens when a solid changes into a liquid.

Example:

ice → liquid water

The temperature at which a substance melts is called its melting point.

Different substances have different melting points.

Boiling

Boiling happens when a liquid changes into a gas throughout the liquid.

The temperature at which a substance boils is called its boiling point.

Evaporation

Evaporation is when a liquid changes into a gas at its surface.

Evaporation can happen below the boiling point.

Practice 1 — Complete the sentences

A substance has different _____.

A solid has a fixed shape and fixed _____.

The temperature at which a solid melt is its _____ point.

The temperature at which a liquid boils is its _____ point.

When a solid changes into a liquid, it is _____.

When a liquid changes into a gas at its surface, it is called _____.

Practice 2 — Choose the correct answer

1. Which is a property of a substance?

- A. Its characteristic
- B. Its name
- C. Its owner
- D. Its location

Answer: _____

2. Which change is melting?

- A. Liquid → solid
- B. Solid → liquid
- C. Liquid → gas
- D. Gas → liquid

Answer: _____

3. Which statements about evaporation are correct?

- A. It only happens at the boiling point.
- B. It happens only to solids.
- C. It happens at the surface of a liquid.
- D. It changes a gas into a liquid.

Answer: _____

Key vocabulary

substance — a type of material

property — something that describes a substance

melting point — temperature at which a solid melt

boiling point — temperature at which a liquid boil

evaporation — change from liquid to gas at the surface

state — solid, liquid or gas

PAGE 2 — THERMAL AND ELECTRICAL CONDUCTORS

1. Thermal conductors

A thermal conductor is a material that allows thermal energy to pass through it easily.

Metals are generally good thermal conductors.

Examples:

Copper, Aluminium, Iron, steel

This is why metal is often used for cooking pans.

2. Thermal insulators

A thermal insulator does not allow thermal energy to pass through it easily.

Examples include:

Plastic, wood, rubber, polystyrene

Insulators can help keep things hot or cold.

Example

A saucepan may have a metal base because metal conducts thermal energy well.

The handle may be made from rubber because rubber is a thermal insulator.

3. Electrical conductors

An electrical conductor allows electricity to pass through it easily.

Many metals are good electrical conductors.

Examples: Copper, Aluminium, iron

Copper is commonly used inside electrical wires because it is affordable.

4. Electrical insulators

An electrical insulator does not allow electricity to pass through it easily.

Examples: Plastic, rubber, wood

Electrical wires often have a metal core covered by plastic.

Why?

The metal allows electricity to flow.

The plastic helps protect people from the electrical current.

Practice 1 — Sort the materials

Write each material in the correct group.

copper | rubber | plastic | Aluminium | wood | iron

Electrical conductors

Electrical insulators

Practice 2 — Choose the best answer

1. Which material is a good electrical conductor?

- A. Rubber**
- B. Plastic**
- C. Copper**
- D. Wood**

Answer: _____

2. Which material is a good thermal insulator?

- A. Copper**
- B. Aluminium**
- C. Plastic**
- D. Iron**

Answer: _____

3. Why can a saucepan have a metal base?

- A. Metal is a good thermal conductor.**
- B. Metal is always soft.**
- C. Metal stops all heat.**
- D. Metal cannot change temperature.**

Answer: _____

Practice 3 — Scientific enquiry

A student tests four materials to find out whether they conduct electricity.

The student uses the same battery, wires, and bulb each time.

Material	Does the bulb light?
Copper	Yes
Plastic	No
Aluminium	Yes
Rubber	No

Questions

1. What is the variable being tested?

2. What is the result for copper?

3. Which materials are electrical conductors?

4. Why should the student use the same battery each time?

Key vocabulary

conductor — a material that allows energy or electricity to pass through easily

insulator — a material that does not allow energy or electricity to pass through easily

thermal energy — energy related to heat

electrical conductor — a material that allows electricity to flow

Remember!

Metal → usually a good conductor

Plastic and rubber → usually good insulators

PAGE 3 — REVERSIBLE CHANGES

1. What is a reversible change?

A reversible change is a change that can be reversed.

The substance can return to its original state or form.

Examples

Ice → water → ice

Melting and freezing are reversible changes.

Water → water vapor → water

Evaporation and condensation can be reversed.

Chocolate → melted chocolate → solid chocolate

The chocolate can melt and then become solid again when it cools.

2. Changes of state

Many changes of state are reversible.

Melting

solid → liquid

Freezing

liquid → solid

Evaporation

liquid → gas

Condensation

gas → liquid

3. Reversible or irreversible?

Ask yourself:

Can I change it back to what it was before?

If yes, it may be a reversible change.

Reversible

melting ice, freezing water, melting chocolate, dissolving some substances, folding paper

Practice 1 — Classify the changes

Write reversible or irreversible.

Ice melting: _____

Water freezing: _____

Chocolate melting: _____

Paper being folded: _____

A substance forming a new substance: _____

Practice 2 — Complete the changes

Solid → liquid = _____

Liquid → solid = _____

Liquid → gas = _____

Gas → liquid = _____

Practice 3 — Explain

1. Why is melting ice a reversible change?

2. A student says, "Evaporation is irreversible because the water disappears." Is the student correct? Explain.

Scientific enquiry

A student wants to investigate whether chocolate can be changed back after melting.

They place chocolate in different places and observe what happens.

Questions

1. What should the student observe?

2. What could they measure?

3. What should they keep the same to make it a fair test?

4. What evidence would show that the change is reversible?

Key vocabulary

reversible change — a change that can be changed back

melting — solid → liquid

freezing — liquid → solid

evaporation — liquid → gas at the surface

condensation — gas → liquid

fair test — a test where only the variable being investigated is changed

PAGE 4 — UNIT 2 PRACTICE TEST

Score: _____ / _____

Part A — Vocabulary

Complete the sentences.

1. A material that allows electricity to pass through easily is an electrical _____.

2. A material that does not allow electricity to pass through easily is an electrical _____.

3. The temperature at which a solid change to a liquid is its _____.

4. A change that can be changed back is called a _____.

Part B — Multiple Choice

5. Which material is usually a good electrical conductor?

- A. Rubber**
- B. Plastic**
- C. Copper**
- D. Wood**

Answer: _____

6. Which change is reversible?

- A. Melting ice**
- B. Burning paper**
- C. Cooking an egg**
- D. Rusting iron**

Answer: _____

7. Which change happens when a liquid becomes a gas at its surface?

- A. Freezing**
- B. Condensation**
- C. Melting**
- D. Evaporation**

Answer: _____

8. Why are some saucepan handles made from plastic?

- A. Plastic is a good thermal insulator.**
- B. Plastic is a good electrical conductor.**
- C. Plastic melts at every temperature.**
- D. Plastic is always a thermal conductor.**

Answer: _____

Part C — Apply Your Knowledge

**9. A student puts a metal spoon and a wooden spoon into hot water.
Which spoon would usually conduct thermal energy better?**

Why?

10. A student has four materials.

Material	Bulb lights?
A	Yes
B	No
C	Yes
D	No

Which materials are electrical conductors?

Part D — Scientific Enquiry

A student wants to find out which material is the best electrical conductor.

They have:

Copper, aluminium, plastic, rubber, a battery, wires, a bulb

11. What is the variable the student should change?

12. What should the student observe or measure?

13. Name ONE thing that should be kept the same.

14. Why is it important to keep this variable the same?

15. The bulb lights when copper is tested but does not light when plastic is tested.

What can the student conclude?

UNIT 2 QUICK REVISION CHECKLIST

Before your test, make sure you can:

- ☐ **Explain what a property is.**
- ☐ **Name the three common states of matter.**
- ☐ **Explain melting and boiling.**
- ☐ **Explain the difference between boiling and evaporation.**
- ☐ **Explain what a thermal conductor is.**
- ☐ **Explain what a thermal insulator is.**
- ☐ **Explain what an electrical conductor is.**
- ☐ **Explain what an electrical insulator is.**
- ☐ **Give examples of conductors and insulators.**
- ☐ **Explain what a reversible change is.**
- ☐ **Identify changes of state.**
- ☐ **Explain melting, freezing, evaporation and condensation.**
- ☐ **Identify variables in a scientific investigation.**
- ☐ **Identify what should be kept the same in a fair test.**

☐ **Use evidence to write a scientific conclusion.**

Important Science Words

**substance • property • solid • liquid • gas • melting • boiling •
evaporation • condensation • freezing • conductor • insulator •
thermal energy • electrical conductor • reversible change • fair test
• variable • evidence • conclusion**